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Academic Details

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Lecturer's Name:	**Mr. T Mavetera**
Group (if applicable)	**N/A**

Assessment Details

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Student Details (Individual, applicable)

Student Full Name:	**Mikaylen Naidoo**
Student Number:	**CON-1459393-V9B6**

Academic Integrity

AI Disclosure Completed	X	**Yes**	**No**

Declaration:
I declare that this assessment is my own original work except for source material explicitly acknowledged. I confirm that this work has not previously been submitted for assessments in any course. I acknowledge the institutional AI policy and confirm that AI tools have only been used as stipulated and documented in the AI Disclosure Appendix.

Student Signature **Mikaylen Naidoo MN**

Date **13 November 2026**

Lecturer Use Only

Lecturer Comments:	
Marks Awarded (%):	
Lecturer Signature:	
Date:	

Assessment:

GITHUB Link to Repository: https://github.com/Mikaylen-Naidoo/CON-1459393-V9B6-ITNPA-Project-1-

ITNPA Project 1: Project Report

Introduction and Objectives

Researchly is a data science research analytics service launched by SAnalytics to help researchers analyse research papers quickly and reduce the time spent screening for relevant articles. Traditional keyword search finds documents that contain specific words, but it does not understand what a paper is about. Automated text categorisation, in which a model learns from labelled documents how to assign new ones to categories, is the standard machine learning response to this problem (Sebastiani, 2002). This project investigates machine learning and text mining approaches that classify and organise research documents by meaning and context.

The project had four parts:

1. **Question 1:** Load a dataset of research abstracts, clean the text, identify useful features and build a baseline classification model.
2. **Question 2:** Train and compare further models, apply one additional performance enhancement, and deploy the best configuration as a web interface.
3. **Question 3:** Apply summarisation techniques to a small collection of full research articles.
4. **Question 4:** Apply Latent Dirichlet Allocation (LDA) topic modelling to uncover the main themes in the same collection and package the whole project in a GitHub repository.

All work was carried out in Python using Jupyter notebooks. The main libraries were pandas for data handling (McKinney, 2010), NLTK for language processing (Loper and Bird, 2002), scikit-learn for machine learning (Pedregosa et al., 2011), gensim and pyLDAvis for topic modelling, and Streamlit for the web interface. The analysis was split across two notebooks: one covering Questions 1 and 2, and one covering Questions 3 and 4.

Question 1: Text Classification Baseline

Data loading and exploration (Question 1.1)

The dataset, research_abstracts.json, was loaded into a pandas DataFrame. It contained 1,138 records and three columns: a DOI (a unique publication identifier), the abstract text, and a label. The label is binary: each abstract is tagged either classical (traditional statistical or econometric research) or ai (research using artificial intelligence or machine learning methods).

Initial profiling showed three characteristics that shaped the rest of the work:

- **Class imbalance.** There were 900 classical abstracts and 238 AI abstracts, roughly an 80 to 20 split. Imbalanced class distributions are a well-documented difficulty for learning algorithms and for model evaluation (He and Garcia, 2009).
- **Missing values.** Four DOIs were missing. No abstracts were missing.
- **Duplicates.** Thirteen abstracts appeared more than once. These were removed, leaving 1,125 records. This step matters because a duplicate can fall in both the training and test sets, which would make a model appear better than it is.

Text denoising

Raw abstracts contain material that adds no meaning for a classifier. A cleaning function was written using NLTK and applied to every abstract, with the result stored in a new column so that the original text was preserved. The steps were:

1. Conversion to lower case, so that "Model" and "model" count as the same word.
2. Removal of HTML tags and web addresses.
3. Removal of mathematical notation written in LaTeX format.
4. Removal of digits, punctuation and symbols, keeping letters only.
5. Removal of English stop words (very common words such as "the" and "of") and tokens of two characters or fewer.

6. Lemmatisation, which reduces words to their base form (for example "networks" becomes "network") so that variants are treated as one term.

Stop word removal and lemmatisation are standard preprocessing steps for reducing vocabulary size and noise in text collections (Manning, Raghavan and Schütze, 2008). The effect here was substantial. The average abstract fell from roughly 171 words to roughly 105 words, meaning around 40% of the original tokens were noise. Charts of the class distribution and the cleaned abstract length by class were produced. They confirmed the imbalance and showed that abstract length overlaps heavily between the two classes, so length alone cannot separate them.

Feature engineering (Question 1.2)

Each column was assessed for its relevance:

- **Abstract:** the only column with real semantic content, and therefore the source of the features.
- **Label:** the target variable the model must predict.
- **DOI:** deliberately excluded. It is almost entirely unique (1,123 distinct values in 1,125 records), so it works as an identifier rather than a source of general patterns. Its prefix identifies the publisher, and a check showed some correlation with the label. For example, IEEE (prefix 10.1109) contributed 44 AI abstracts against 9 classical, while SSRN (prefix 10.2139) contributed 261 classical against 31 AI. A model using this would be learning publishing habits rather than the content of the research, which defeats the purpose of the service. It was therefore excluded to avoid this kind of shortcut.

The data was split into 80% training and 20% testing, stratified so that both parts kept the same class proportions, with a fixed random seed for reproducibility. The split was made before building features so that no information from the test set could leak into the model.

The cleaned text was converted into numbers using TF-IDF (term frequency by inverse document frequency), a term weighting scheme that scores a word highly when it is frequent within one document but rare across the collection (Salton and Buckley, 1988; Manning, Raghavan and Schütze, 2008). Distinctive words therefore carry more weight than generic ones. The vectoriser used single words and two-word phrases, ignored terms appearing in fewer than three abstracts or more than 90% of abstracts, and dampened the effect of very frequent words. It was fitted on the training set only. This produced a matrix of about 900 training abstracts and about 6,600 features.

A chi-square test, a filter method of feature selection, then ranked each feature by its statistical association with the label (Rogati and Yang, 2002). The strongest terms were network, machine, neural, neural network, machine learning, learning, algorithm, LSTM, random forest and SVR. These are exactly the vocabulary expected in AI research, which confirms that the cleaned abstract text is an informative feature source.

Baseline model and performance (Question 1.3)

A logistic regression classifier was chosen as the baseline because linear models work well with the high-dimensional, sparse features typical of text (Joachims, 1998; Hastie, Tibshirani and Friedman, 2009), and because the model is fast and easy to interpret. Class weights were set to "balanced" so that errors on the smaller AI class counted more heavily, a cost-sensitive approach that prevents the model from scoring well simply by favouring the majority class (He and Garcia, 2009).

Dataset	Accuracy	F1 (AI class)	ROC-AUC
Training	0.986	0.966	0.999
Test	0.867	0.694	0.890

On the test set the model classified 161 classical and 34 AI abstracts correctly. It wrongly labelled 17 classical abstracts as AI and missed 13 AI abstracts. For the AI class, precision was about 0.67 and recall about 0.72.

Five-fold cross-validation gave a mean F1 of about 0.76 with a standard deviation of about 0.03. A learning curve showed training F1 staying between about 0.95 and 0.99, while validation F1 rose only slowly from about 0.68 to 0.76. A review of the model weights showed sensible learning: AI terms included network, learning, neural, machine and algorithm, while classical terms included policy, monetary, interest, nominal and test.

Interpretation

- **Overfitting.** The model is near perfect on training data but clearly weaker on unseen data. With about 6,600 features and only about 900 training examples, it can partly memorise the training set. A large gap between training and validation performance, with curves that have not converged, is the typical signature of high variance (Hastie, Tibshirani and Friedman, 2009).
- **Accuracy is misleading.** Because 80% of abstracts are classical, a model that always predicted "classical" would achieve about 80% accuracy. Precision, recall and the F1 score for the AI class are therefore more honest measures (He and Garcia, 2009). The F1 of about 0.69 gives a more realistic picture than the accuracy of 0.87. For the same reason, ROC-AUC should be read with caution on imbalanced data, because it can give an optimistic impression of performance (Saito and Rehmsmeier, 2015).
- **Likely sources of error.** Mistakes probably arise where classical papers mention words such as "algorithm" or "learning" in passing, and where AI papers are written in economics language with little machine learning jargon.
- **More data would help.** Validation performance was still rising as the training set grew.

Question 2: Model Improvement and Deployment

Comparison of models (Question 2.1)

Four further models were trained, each from a different family: a Linear Support Vector Machine (Cortes and Vapnik, 1995), Complement Naive Bayes (Manning, Raghavan and Schütze, 2008), a Random Forest (Breiman, 2001) and Gradient Boosting (Friedman, 2001). Support vector machines were chosen in part because they have been shown to perform strongly on text categorisation tasks (Joachims, 1998). Each model was placed in a pipeline with the same TF-IDF step so that the comparison was fair and the vectoriser was refitted correctly inside each cross-validation fold. All models were evaluated on the same test set using accuracy, precision, recall, F1 for the AI class, ROC-AUC and cross-validated F1.

Model	Accuracy	Precision	Recall	F1 (AI)	ROC-AUC	CV F1
Random Forest	0.907	0.825	0.702	0.759	0.924	0.812
Gradient Boosting	0.898	0.816	0.660	0.729	0.930	0.752
Linear SVM	0.889	0.775	0.660	0.713	0.894	0.823
Logistic Regression (baseline)	0.867	0.667	0.723	0.694	0.890	0.773
Complement Naive Bayes	0.880	0.857	0.511	0.640	0.880	0.699

Random Forest, Gradient Boosting and the Linear SVM all outperformed the baseline on test F1. The Random Forest was the best overall, raising F1 from 0.694 to 0.759 and accuracy from 0.867 to 0.907. Naive Bayes achieved the highest precision (0.857) but very low recall (0.511), so it missed about half of the AI abstracts and finished below the baseline on F1. This weakness motivated the enhancement described next.

It should be noted that the test set contains only 47 AI abstracts, so differences of two to three points in F1 are within the range of random variation. For that reason, cross-validated scores were reported alongside test scores.

Performance enhancement: decision threshold optimisation (Question 2.2)

By default, a classifier labels an abstract as AI when its predicted probability is at least 0.5. With imbalanced classes, that cut-off is often not the best choice, which is one of the practical difficulties of imbalanced learning (He and Garcia, 2009). The enhancement chosen was to find, for each model, the probability threshold that maximises the F1 score for the AI class.

To keep the evaluation honest, the threshold was selected using out-of-fold predictions on the training data only, in line with standard cross-validation practice (Hastie, Tibshirani and Friedman, 2009). The test set was used only for the final before-and-after comparison. For the Support Vector Machine, probabilities were obtained through a calibration wrapper.

	Linear SVM	Complement NB	Random Forest	Gradient Boosting
Selected threshold	0.47	0.33	0.50	0.54
Test F1, default to tuned	0.713 to 0.725	0.640 to 0.688	0.759 to 0.759	0.729 to 0.714
Test recall, default to tuned	0.660 to 0.702	0.511 to 0.681	0.702 to 0.702	0.660 to 0.638
Cross-validated F1, default to tuned	0.815 to 0.822	0.700 to 0.780	0.813 to 0.813	0.752 to 0.759

The results showed where the technique helps:

- **Naive Bayes benefited most.** Lowering its threshold to 0.33 raised AI recall from 0.51 to 0.68, test F1 from 0.640 to 0.688 and cross-validated F1 from 0.70 to 0.78. The model had been under-predicting the minority class, and the threshold corrected this bias.
- **The Linear SVM improved slightly** (test F1 up by about 0.012).
- **The Random Forest was unchanged**, because its best threshold was already 0.50, indicating it was already well balanced.
- **Gradient Boosting** improved in cross-validation but dipped slightly on the small test set. This is the kind of noise expected with only 47 minority-class test examples, and it is reported as observed.

Grid-search hyperparameter tuning was also tried on the SVM and Random Forest, and it did not improve test performance, so it was not adopted as the showcased enhancement. The overall conclusion is that threshold tuning helps models biased towards the majority class and does no harm to well-calibrated ones, but it does not change which model is best.

Deployment as a web interface (Question 2.3)

The best configuration was the Random Forest with TF-IDF features at a threshold of 0.50 (test accuracy 0.907, F1 0.759, ROC-AUC 0.924). The complete pipeline, together with its threshold, was saved to a single model file of about 5.6 MB. Saving the whole pipeline means the application does not need to rebuild the vocabulary separately.
A Streamlit web application was written to match the interface shown in the brief. It presents a title ("Researchly Research Abstract Classifier"), a short description, a text box for the abstract, a "Classify Abstract" button, a "Prediction" heading and a green result box. When an abstract is entered, the app applies the same text cleaning used in training, calculates the model's probability for the AI class, compares it with the saved threshold and displays either "artificial intelligence" or "classical research". It also shows the model's confidence and warns the user if the text box is empty.
The application was tested with the sample abstract from the brief, which describes artificial intelligence in healthcare, and it correctly returned "artificial intelligence". A second test with an economics abstract about monetary policy and interest rates correctly returned "classical research".
Because Streamlit runs as a continuous web server, it cannot finish as an ordinary notebook cell, and running it that way leaves the cell permanently busy. The application is therefore launched from a terminal (Anaconda Prompt) by navigating to the project folder and running the Streamlit command, after which it opens in a browser at http://localhost:8501. The application file and the saved model must be kept in the same folder.

Question 3: Topic Analysis and Document Summarisation

Corpus creation (Question 3.1)
The corpus comprised five full-length articles supplied as text files. Each file was read into a DataFrame row with an identifier and the full content. Their lengths were:

Article	Words
research_1	5,602
research_2	2,416
research_3	3,828
research_4	5,922
research_5	4,027

Inspection of the contents showed that the articles concern research methodology: research_1 on mixed methods research, research_2 on researcher position and reflexivity in qualitative research, research_3 on developmental research in instructional design, research_4 on the use of Cronbach's alpha, and research_5 on triangulation in qualitative research. The brief describes the corpus as AI research articles, so this difference should be noted. It does not affect the validity of the methods, which work on any collection of documents, but it does shape the topics that were found.

Preprocessing (Question 3.2)
A two-stage cleaning process was applied. The first stage, denoising, handled problems typical of text extracted from PDF documents and academic writing: words broken across lines by hyphens were rejoined, in-text citations such as author and year references were removed, along with bracketed numbers, web and email addresses, page references and footnote numbers attached to words. The output of this stage remained as readable sentences, which the summariser requires.
The second stage converted the text to tokens by lowercasing, keeping letters only, lemmatising and removing stop words, using NLTK (Loper and Bird, 2002) and the standard preprocessing approach described by Manning, Raghavan and Schütze (2008). In addition to the standard English stop word list, a custom list of academic terms was removed, including "study", "paper", "article", "journal" and "research". The last of these is deliberate: in a collection about research methods, it appears in nearly every sentence and would overwhelm the real topics. Two artefacts of the lemmatiser ("wa" from "was" and "ha" from "has") were also removed.
This removed roughly half of the tokens. For example, research_1 fell from 5,602 words to 2,614 tokens, and research_4 from 5,922 to 2,929.

Summary of the first article (Question 3.3)
The first article was summarised using a TextRank-style extractive method, meaning it selects the most important existing sentences rather than writing new ones. Each sentence becomes a node in a graph, sentences are connected according to their similarity (measured with TF-IDF and cosine similarity), and a PageRank algorithm (Brin and Page, 1998) scores each sentence by how central it is to the others. This follows the graph-based sentence centrality idea of LexRank (Erkan and Radev, 2004). The five highest scoring sentences were returned in their original order.
During testing, the first attempt returned repeated sentences, because the source file contained duplicated passages (156 sentences before and 81 after removing repeats). Repeated sentences were therefore removed before scoring, and footnote markers attached to words were stripped. The final summary was about 3% of the original length and captured the three core messages of the article: the aim of positioning mixed methods research as the third research paradigm alongside quantitative and qualitative research, the "purist" view that the two paradigms are incompatible, and the argument that pragmatism provides a philosophical basis for combining them.
Three visualisations supported the summary: a bar chart of the top sentence scores, a bar chart of the fifteen most frequent terms (method with about 80 mentions, followed by mixed, qualitative, quantitative, paradigm, purist and reality), and a word cloud.

Aggregate summary of the remaining articles (Question 3.4)
A different technique was used for the other four articles: TF-IDF keyword and key phrase extraction (Salton and Buckley, 1988). Rather than selecting sentences, it identifies the terms that distinguish each article from the others. Terms appearing in more than half of the articles were excluded so that generic vocabulary did not dominate, and a term's average score across the four articles gave an aggregate ranking.

Article	Highest scoring terms	Theme
research_2	reflexivity, woman, immigrant, participant, personal	Researcher position and reflexivity
research_3	developmental, instructional, design, development process	Developmental research in instructional design
research_4	alpha, Cronbach, scale, item, statistic	Use and interpretation of Cronbach's alpha
research_5	Denzin, triangulation, data triangulation, mixed method	Triangulation in qualitative research

The aggregate ranking was led by design, development, participant, reflexivity, personal, bias, reliability and triangulation. Three visualisations were produced: bar charts of the top terms for each article, a

heatmap of the leading aggregate terms across the articles, and an aggregate word cloud. The heatmap showed that each article has a distinctive vocabulary (for example reflexivity appears only in research_2 and alpha only in research_4), while some terms link articles, such as bias and participant across research_2 and research_5. Taken together, the corpus is about the quality, rigour and design of research methodology.

Question 4: Topic Modelling with LDA

Preparing the corpus (Question 4.1)
Latent Dirichlet Allocation is a generative probabilistic model in which each document is a mixture of topics and each topic is a distribution over words (Blei, Ng and Jordan, 2003). It has been widely applied to discover themes in collections of scientific abstracts and papers (Griffiths and Steyvers, 2004). LDA needs many documents to learn word patterns, and five long articles are too few. Each article was therefore divided into passages of about 150 words, with fragments of fewer than 50 words discarded. This produced 140 passages (36 from research_1, 15 from research_2, 25 from research_3, 39 from research_4 and 25 from research_5), each linked to its source article. The passages were tokenised using the same cleaning as in Question 3.

A dictionary mapping words to numeric identifiers was built, and words occurring in fewer than three passages (too rare to model) or more than 40% of passages (too common to be informative) were removed. This left a vocabulary of about 800 words. Each passage was then converted to a bag-of-words representation.

Choosing the number of topics
LDA requires the number of topics to be chosen in advance. Models with two to eight topics were trained and compared using c_v coherence, which measures how semantically related the leading words of each topic are and correlates well with human judgements of topic interpretability (Röder, Both and Hinneburg, 2015). Because LDA involves randomness and the corpus is small, each score was averaged over three random seeds.

Topics	2	3	4	5	6	7	8
Mean coherence	0.494	0.518	0.557	0.471	0.499	0.474	0.500

Coherence peaked at four topics, so four were used. The final model had a coherence of 0.576.

The topics found

Topic	Leading words	Interpretation
0	alpha, instrument, value, Cronbach, science, reliability, scale, item	Reliability and Cronbach's alpha
1	data, triangulation, Denzin, method, multiple, source, methodological	Triangulation and multiple data sources
2	method, qualitative, mixed, design, quantitative, paradigm, reflexivity, position	Mixed methods and the qualitative and quantitative paradigms
3	development, process, developmental, design, instructional, evaluation	Instructional and developmental research design

The names given to the topics are interpretations of the word lists. Bar charts showed the word probabilities within each topic. The proportion of passages whose dominant topic was each of the four was reasonably balanced: 29%, 17%, 28% and 26%.

How the topics map to the articles
The average topic mixture was calculated for each article and displayed as a heatmap and stacked bar chart.

Article	Main topic	Share
research_4 (Cronbach's alpha)	Topic 0	98%
research_3 (developmental research)	Topic 3	91%
research_5 (triangulation)	Topic 1	55%, with 41% in Topic 2
research_2 (reflexivity)	Topic 2	61%
research_1 (mixed methods)	Topic 2	48%, with 30% in Topic 3

This is a strong validation of the model. It recovered the theme of each article without ever being told which passage came from which article. The overlap is also sensible: triangulation and reflexivity both belong to

the broader family of qualitative and mixed methods thinking. One caveat is that, with only five articles, the topics are sensitive to the random starting point. This is why the random seed was fixed and coherence was averaged across several seeds.

Interactive visualisation (Question 4.2)

An interactive visualisation was produced with pyLDAvis, a Python implementation of the LDAvis approach (Sievert and Shirley, 2014), and saved as a standalone HTML file. The left panel is the Intertopic Distance Map, where each circle is a topic. Circle size reflects how much of the corpus the topic covers, and the distance between circles reflects how different the topics are. The right panel lists the most salient words for the selected topic, with a relevance slider that moves between words that are frequent overall and words that are distinctive to the topic. Sievert and Shirley (2014) suggest a setting around 0.6 as a useful compromise.
During testing, the default projection method failed with a numerical error caused by a newer version of the numerical library. This was resolved by selecting an alternative projection method (multidimensional scaling on a metric basis), which produced a valid map.

Question 4.3: Repository, Documentation and Reproducibility

The project is packaged in a GitHub repository with the following structure:

Folder	Contents
docs/	This documentation, covering Questions 1 to 4
data/	The datasets: research_abstracts.json and the five article text files
notebooks/	The two Jupyter notebooks showing the flow of analytics for Questions 1 to 4
outputs/	Generated graphs, application screenshots and the LDA visualisation
deployment/	The Streamlit application, saved model, requirements file and a README with deployment steps

The main README summarises the project and links to each part, and the deployment README gives step by step instructions: install the requirements, keep the application and model files together, launch the application with Streamlit and open the local address in a browser. To run the analysis, each notebook is run from top to bottom, which reproduces every result and graph. The notebooks contain saved outputs so that results can be read without re-running the code.

Conclusion

The project delivered a complete text analytics workflow. A cleaned dataset of 1,125 abstracts supported a baseline classifier with a test F1 of 0.694. Comparing four further models and applying threshold optimisation showed that a Random Forest performed best, reaching an accuracy of 0.907 and an F1 of 0.759, and this model was deployed as a working web interface. Summarisation of the article collection produced a concise extractive summary of one article and an aggregate keyword summary of the other four. LDA topic modelling identified four coherent themes that corresponded closely to the content of each article, and an interactive map allowed the topics to be explored. Together, these results show that machine learning and text mining can classify and organise research documents by meaning, supporting the aims of the Researchly service.

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AI Evidence Section

[AI Code 1 – 4 may necessitate evidence submission as specified in your assessment instruction, this section is reserved for evidence where relevant, please delete if not applicable.]

AI Disclosure Appendix

AI tool used	Purpose / how it was used	How I verified or adapted the output
Claude Sonnet 5.5	Brainstormed initial argument outline	Cross-checked with the peer-reviewed sources; fully rewritten and adapted by me, I take full responsibility for the final submission
Grammarly	Grammer and plagiarism checker	Helped me identify gaps in spelling and grammar, fixed minor issues work is still my own, used as the plagiarism checker

Declaration:

I/we declare that this assessment is my/our own original work, except for source material explicitly acknowledged. I/we confirm that any use of Artificial Intelligence (AI) tools has been applied only as permitted by the assessment instructions and has been fully disclosed where required.

I/we understand that I/we remain responsible for the accuracy, originality, and academic quality of this submission. I/we am/are aware of the institution's policies on academic integrity and the requirements relating to the responsible use of AI in assessments.

I/we acknowledge that failure to comply with these requirements, including undeclared or inappropriate AI use, may result in academic penalties or disciplinary procedures in accordance with institutional policy.

Student Signature(s)	Date
MIKAYLEN NAIDOO	**11 November 2026**